The Ploughman family owns and operates a 640-acre farm that has been in the family for several generations. The Ploughmans always have had to work hard to make a decent living from the farm and have had to endure some occasional difficult years. Stories about earlier generations overcoming hardships due to droughts, floods, and so forth, are an important part of the family history. However, the Ploughmans enjoy their self-reliant lifestyle and gain considerable satisfaction from continuing the family tradition of successfully living off the land during an era when many family farms are being abandoned or taken over by large agricultural corporations.

John Ploughman is the current manager of the farm, while his wife Eunice runs the house and manages the farm's finances. John's father, Grandpa Ploughman, lives with them and still puts in many hours working on the farm. John and Eunice's older children, Frank, Phyllis, and Carl, also are given heavy chores before and after school.

The entire family can produce a total of 4,000 person-hours' worth of labor during the winter and spring months and 4,500 person-hours during the summer and fall. If any of these person hours are not needed, Frank, Phyllis, and Carl will use them to work on a neighboring farm for $5/hour during the winter and spring months and $5.50/hour during the summer and fall.

The farm supports two types of livestock, dairy cows and laying hens, as well as three crops: soybeans, corn, and wheat. (All three are cash crops, but the corn also is a feed crop for the cows and the wheat also is used for chicken feed.) The crops are harvested during the late summer and fall. During the winter months, John, Eunice, and Grandpa decide about the mix of livestock and crops for the coming year.

Currently, the family has just completed a particularly successful harvest that has provided an investment fund of $20,000 that can be used to purchase more livestock. (Other money is available for ongoing expenses, including the next planting of crops.) The family currently has 30 cows valued at $35,000 and 2,000 hens valued at $5,000. They wish to keep all this livestock and perhaps purchase more. Each new cow would cost $1,500, and each new hen would cost $3.

Over a year's time, the value of a herd of cows will decrease by about 10 percent and the value of a flock of hens will decrease by about 25 percent due to aging.
Each cow will require two acres of land for grazing and 10 person-hours of work per month, while producing a net annual cash income of $850 for the family. The corresponding figures for each hen are no significant acreage, 0.05 person-hours per month, and an annual net cash income of $4.25.

The chicken house can accommodate a maximum of 5,000 hens, and the size of the barn limits the herd to a maximum of 42 cows.

For each acre planted in each of the three crops, the next table gives the number of person-hours of work that will be required during the first and second halves of the year, as well as a rough estimate of the crop's net value (in either income or savings in purchasing feed for the livestock).

Data per Acre Planted

	Soybeans	Corn	Wheat
Winter and spring, person-hours	1.0	0.9	0.6
Summer and fall, person-hours	1.4	1.2	0.7
Net value	$70	$60	$40

To provide much of the feed for the livestock, John wants to plant at least one acre of corn for each cow in the coming year's herd and at least 0.05 acre of wheat for each hen in the coming year's flock.

John, Eunice, and Grandpa now are discussing how much acreage should be planted in each of the crops and how many cows and hens to have for the coming year. Their objective is to maximize the family's monetary worth at the end of the coming year (**the sum of the net income from the livestock for the coming year** + **the net value of the crops for the coming year** + **what remains from the investment fund** + **the value of the livestock at the end of the coming year** + **income from working on a neighboring farm - living expenses of $40,000 for the year**).

a. **Identify verbally the components of a linear programming model for this problem.**

Decision Variables: The decision variables are the choices that the Ploughman family must make to optimize their situation.

- C: The number of cows in the herd.
- H: The number of hens in the flock.
- A1: The number of acres to plant in soybeans.

- A2: The number of acres to plant in corn.
- A3: The number of acres to plant in wheat.

Objective Function: The objective function is to maximize their monetary worth, which can be broken down into the following components:

- **Net Income from Livestock**

Each cow provides a net income of $850 per year, Income from cows is **850 × C**

Each hen provides a net income of $4.25 per year, Income from hens is **4.25 × H**

- **Net Value of Crops**

Soybeans generate a net value of $70 per acre, Value from soybeans is **70 × A1**

Corn generates a net value of $60 per acre, Value from corn is **60 × A2**

Wheat generates a net value of $40 per acre, Value from wheat is **40 × A3**

- **Remaining Investment Fund**

The remaining investment fund after purchasing livestock is the amount left from the **$20,000**, considering the costs of cows and hens.

Each Cows cost **$1,500**

Each Hens cost **$3**

So, the total cost of livestock is **1500 × C + 3 × H**

The remaining investment fund is **20,000 - (1,500 × C + 3 × H)**

- **Value of Livestock at the End of the Year**

The value of the herd of cows decreases by 10% annually
Value of each cow at the end of the year is **90% of $1,500 = $1,350**

The value of the flock of hen decreases by 25% annually
So, the value of each hen at the end of the year is **75% of $3 = $2.25**

Thus, the total value of livestock at the end of the year is **1,350 × C + 2.25 × H**

- **Income from Off-Farm Work**

The children can earn **$5/hour** during winter/spring and **$5.50/hour** during summer/fall.

- **Living Expenses**

The family's living expenses for the year are **$40,000**

Objective function is:

$$\text{Maximize } Z = (850 * C) + (4.25 * H) + (70 * A1) + (60 * A2) + (40 * A3) + (20{,}000 - (1{,}500 * C + 3 * H)) + (1350 * C) + (2.25 * H) - 40000$$

Constraints: The constraints represent the restrictions on labor, land, feed requirements, and other factors.

a. Land Constraints

Each cow requires 2 acres for grazing, so the total land required is **2 * C**

Soybeans, corn, and wheat require the respective acres **A1, A2**, and **A3**

The total land constraint is: **(2 * C) + A1 + A2 + A3 ≤ 640**

b. Labor Constraints

4,000 person-hours during winter/spring

4,500 person-hours during summer/fall

The person-hours required for each crop per acre are given in the table, as well as for each cow and hen:

Cows: 10 person-hours per month for each cow (120 hours per year per cow)

Hens: 0.05 person-hours per month for each hen (0.6 hours per year per hen)

Data per Acre Planted

	Soybeans	Corn	Wheat
Winter and spring, person-hours	1.0	0.9	0.6
Summer and fall, person-hours	1.4	1.2	0.7
Net value	$70	$60	$40

The labor constraints are:

Winter/Spring labor: **1 * A1 + 0.9 * A2 + 0.6 * A3 + 120 * C + 0.6 * H ≤ 4000**

Summer/Fall labor: **1.4 * A1 + 1.2 * A2 + 0.7 * A3 + 120 * C + 0.6 * H ≤ 4500**

c. Feed Constraints:

The number of acres of corn must be at least equal to the number of cows → **A2 ≥ C**

The number of acres of wheat must be at least 0.05 acres per hen → **A3 ≥ 0.05 * H**

d. Livestock Constraints

Cows: The maximum number of cows is 42 (Barn Space Constraint)

Hens: The maximum number of hens is 5,000 (Chicken House Capacity)

These constraints are: **C ≤ 42 and H ≤ 5000**

Non-Negativity Constraints: All decision variables must be **non-negative** because the family cannot have negative numbers of cows, hens, or acres of crops

C ≥ 0, H ≥ 0, A1 ≥ 0, A2 ≥ 0, A3 ≥ 0

b. Display the model on a spreadsheet.

Plantings				Planting		
	Soybeans	Corn	Wheat	Totals		
W&S Hours Required	1	0.9	0.6	537		
S&F Hours Required	1.4	1.2	0.7	736		
Net Value	$70	$60	$40	$37,300		
Acres Planted	450	30	100	580		
		>=	>=			
		30	100			
		1	0.05			
		acre/cow	acre/hen			
Livestock			Livestock			
	Cows	Hens	Totals			
Hours Required per Month	10	0.05	400			
Grazing Land Required	2	0	60			
Net Annual Cash Income	$850	$4.25	$34,000			
ning Value (Current Livestock)	$35,000	$5,000				
Decrease in Value per Year	10%	25%				
End Value (Current Livestock)	$31,500	$3,750	$35,250		Investment	
					Fund	
Cost of New Livestock	$1,500	$3.00	$0	<=	$20,000	
End Value (New Livestock)	$1,350	$2.25	$0			
Current Livestock	30	2,000				
New Livestock	0	0				
Total Livestock	30	2,000				
	<=	<=				
Barn/House Limits	42	5,000				
Neighboring Farm Work			Neighbor			
	W&S	S&F	Totals			
Wage	$5	$5.50	$12,817			
Hours Worked	1,063	1,364	2427			
Totals	Plantings	Livestock	Neighbor	Total		Available
W&S Hours	537	2,400	1,063	4,000	<=	4,000
S&F Hours	736	2,400	1,364	4,500	<=	4,500
Acreage	580	60	0	640	<=	640
Net Income	$37,300	$34,000	$12,817	$84,117		
End of Year Value		$35,250		$35,250		
Leftover Investment Fund		$20,000		$20,000		
Living Expenses				$40,000		
Total Monetary Worth				$99,367		

c. Obtain an optimal solution and generate the sensitivity report. What does the model predict regarding the family's monetary worth at the end of the coming year?

Microsoft Excel 16.96 Sensitivity Report
Worksheet: [Case Template.xlsx]Case 5-3b
Report Created: 4/25/25 19:51:37

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
B27	New Livestock Cows	0	-53	70	53	1E+30
C27	New Livestock Hens	0	-0.8575	0.35	0.8575	1E+30
B7	Acres Planted Soybeans	450	0	57.3	1E+30	8.4
C7	Acres Planted Corn	30	0	48.9	8.4	1E+30
D7	Acres Planted Wheat	100	0	33.15	17.15	1E+30

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
B28	Total Livestock Cows	30	0	42	1E+30	12
C28	Total Livestock Hens	2000	0	5000	1E+30	3000
C7	Acres Planted Corn	30	-8.4	0	450	30
D23	Cost of New Livestock Totals	$0.00	0	20000	1E+30	20000
D7	Acres Planted Wheat	100	-24.15	0	450	100
E39	W&S Hours Total	4000	0	4000	1E+30	0
E40	S&F Hours Total	4500	0	4500	1E+30	0
E41	Acreage Total	640	57.3	640	1E+30	450

d. Find the allowable range for the net value per acre planted for each of the three crops.

From the Solver sensitivity report, we can extract the allowable range for the net value per acre planted for each crop from the Objective Coefficient, Allowable Increase, and Allowable Decrease columns.

Allowable Range for Net Value per Acre (Crop Profitability)

Crop	Current Net Value	Allowable Increase	Allowable Decrease	Range of Values
Soybeans	$57.30	∞ (1E+30)	$8.40	$48.90 to infinity
Corn	$48.90	$8.40	∞ (1E+30)	$48.90 to $57.30
Wheat	$33.15	$17.15	∞ (1E+30)	$33.15 to $50.30

Interpretation:

- **Soybeans**: Profitable as-is; even a decrease of up to $8.40/acre still keeps it in the optimal solution.
- **Corn and Wheat**: Just meeting feed constraints; their current net values are right at the break-even. Any lower, and they would be dropped from the solution unless required for livestock.

Getting no new livestock is the optimal solution because It maximizes your total monetary worth by avoiding costly investments that don't generate higher returns than neighbor farm work.

High cost	Cows cost $1,500 upfront — that money is more valuable kept
Depreciation	You lose 10% (cows) and 25% (hens) of value within a year

Added labor & land	More animals → more time, feed, grazing land needed
Better alternative	Extra labor is better used earning $5–$5.5/hr working off-farm

The above estimates of the net value per acre planted in each of the three crops assumes good weather conditions. Adverse weather conditions would harm the crops and greatly reduce the resulting value. The scenarios particularly feared by the family are a drought, a flood, an early frost, both a drought and an early frost, and both a flood and an early frost. The estimated net values for the year under these scenarios are shown next.

	Net Value per Acre Planted		
Scenario	**Soybeans**	**Corn**	**Wheat**
Drought	−$10	−$15	0
Flood	15	20	$10
Early frost	50	40	30
Drought and early frost	−15	−20	−10
Flood and early frost	10	10	5

e. **Find an optimal solution under each scenario after making the necessary adjustments to the linear programming model formulated in part b. In each case, what is the prediction regarding the family's monetary worth at the end of the year?**

Drought

Plantings				Planting		
	Soybeans	Corn	Wheat	Totals		
W&S Hours Required	1	0.9	0.6	117.6		
S&F Hours Required	1.4	1.2	0.7	143.5		
Net Value	-$10	-$15	$0	-$630		
Acres Planted	0	42	133	175		
		>=	>=			
		42	133			
		1	0.05			
		acre/cow	acre/hen			
Livestock			Livestock			
	Cows	Hens	Totals			
Hours Required per Month	10	0.05	553			
Grazing Land Required	2	0	84			
Net Annual Cash Income	$850	$4.25	$47,005			
ng Value (Current Livestock)	$35,000	$5,000				
Decrease in Value per Year	10%	25%				
nd Value (Current Livestock)	$31,500	$3,750	$35,250		Investment	
					Fund	
Cost of New Livestock	$1,500	$3.00	$19,980	<=	$20,000	
End Value (New Livestock)	$1,350	$2.25	$17,685			
Current Livestock	30	2,000				
New Livestock	12	660				
Total Livestock	42	2,660				
	<=	<=				
Barn/House Limits	42	5,000				
Neighboring Farm Work			Neighbor			
	W&S	S&F	Totals			
Wage	$5	$5.50	$8,534			
Hours Worked	564	1,039	1602.9			
Totals	Plantings	Livestocl	Neighbor	Total		Available
W&S Hours	117.6	3,318	564	4,000	<=	4,000
S&F Hours	143.5	3,318	1,039	4,500	<=	4,500
Acreage	175	84	0	259	<=	640
Net Income	-$630	$47,005	$8,534	$54,909		
End of Year Value		$52,935		$52,935		
Leftover Investment Fund		$20		$20		
Living Expenses				$40,000		
Total Monetary Worth				$67,864		

Range Name	Cells
AcresPlanted	B7:D7
Available	G39:G41
BarnHouseLimits	B30:C30
CurrentLivestock	B26:C26
HoursWorked	B36:C36
InvestmentFund	F23
MonetaryWorth	E47
NewLivestock	B27:C27
Total	E39:E41
TotalLivestock	B28:C28
Wage	B34:C34

Flood

Plantings				Planting		
	Soybeans	Corn	Wheat	Totals		
W&S Hours Required	1	0.9	0.6	460.5		
S&F Hours Required	1.4	1.2	0.7	600.7		
Net Value	$15	$20	$10	$9,790		
Acres Planted	0	423	133	556		
		>=	>=			
		42	133			
		1	0.05			
		acre/cow	acre/hen			
Livestock			Livestock			
	Cows	Hens	Totals			
Hours Required per Month	10	0.05	553			
Grazing Land Required	2	0	84			
Net Annual Cash Income	$850	$4.25	$47,005			
ning Value (Current Livestock)	$35,000	$5,000				
Decrease in Value per Year	10%	25%				
End Value (Current Livestock)	$31,500	$3,750	$35,250		Investment	
					Fund	
Cost of New Livestock	$1,500	$3.00	$19,980	<=	$20,000	
End Value (New Livestock)	$1,350	$2.25	$17,685			
Current Livestock	30	2,000				
New Livestock	12	660				
Total Livestock	42	2,660				
	<=	<=				
Barn/House Limits	42	5,000				
Neighboring Farm Work			Neighbor			
	W&S	S&F	Totals			
Wage	$5	$5.50	$4,305			
Hours Worked	222	581	802.8			
Totals	Plantings	Livestock	Neighbor	Total		Available
W&S Hours	460.5	3,318	222	4,000	<=	4,000
S&F Hours	600.7	3,318	581	4,500	<=	4,500
Acreage	556	84	0	640	<=	640
Net Income	$9,790	$47,005	$4,305	$61,100		
End of Year Value		$52,935		$52,935		
Leftover Investment Fund		$20		$20		
Living Expenses				$40,000		
Total Monetary Worth				$74,055		

Range Name	Cells
AcresPlanted	B7:D7
Available	G39:G41
BarnHouseLimits	B30:C30
CurrentLivestock	B26:C26
HoursWorked	B36:C36
InvestmentFund	F23
MonetaryWorth	E47
NewLivestock	B27:C27
Total	E39:E41
TotalLivestock	B28:C28
Wage	B34:C34

Early Frost

Plantings				Planting		
	Soybeans	Corn	Wheat	Totals		
W&S Hours Required	1	0.9	0.6	537		
S&F Hours Required	1.4	1.2	0.7	736		
Net Value	$50	$40	$30	$26,700		
Acres Planted	450	30	100	580		
		>=	>=			
		30	100			
		1	0.05			
		acre/cow	acre/hen			
Livestock			Livestock			
	Cows	Hens	Totals			
Hours Required per Month	10	0.05	400			
Grazing Land Required	2	0	60			
Net Annual Cash Income	$850	$4.25	$34,000			
ing Value (Current Livestock)	$35,000	$5,000				
Decrease in Value per Year	10%	25%				
End Value (Current Livestock)	$31,500	$3,750	$35,250		Investment	
					Fund	
Cost of New Livestock	$1,500	$3.00	$0	<=	$20,000	
End Value (New Livestock)	$1,350	$2.25	$0			
Current Livestock	30	2,000				
New Livestock	0	0				
Total Livestock	30	2,000				
	<=	<=				
Barn/House Limits	42	5,000				
Neighboring Farm Work			Neighbor			
	W&S	S&F	Totals			
Wage	$5	$5.50	$12,817			
Hours Worked	1,063	1,364	2427			
Totals	Plantings	Livestock	Neighbor	Total		Available
W&S Hours	537	2,400	1,063	4,000	<=	4,000
S&F Hours	736	2,400	1,364	4,500	<=	4,500
Acreage	580	60	0	640	<=	640
Net Income	$26,700	$34,000	$12,817	$73,517		
End of Year Value		$35,250		$35,250		
Leftover Investment Fund		$20,000		$20,000		
Living Expenses				$40,000		
Total Monetary Worth				$88,767		

Range Name	Cells
AcresPlanted	B7:D7
Available	G39:G41
BarnHouseLimits	B30:C30
CurrentLivestock	B26:C26
HoursWorked	B36:C36
InvestmentFund	F23
MonetaryWorth	E47
NewLivestock	B27:C27
Total	E39:E41
TotalLivestock	B28:C28
Wage	B34:C34

Drought and Early Frost

Plantings				Planting		
	Soybeans	Corn	Wheat	Totals		
W&S Hours Required	1	0.9	0.6	97.8		
S&F Hours Required	1.4	1.2	0.7	120.4		
Net Value	-$15	-$20	-$10	-$1,840		
Acres Planted	0	42	100	142		
		>=	>=			
		42	100			
		1	0.05			
		acre/cow	acre/hen			
Livestock			Livestock			
	Cows	Hens	Totals			
Hours Required per Month	10	0.05	520			
Grazing Land Required	2	0	84			
Net Annual Cash Income	$850	$4.25	$44,200			
ng Value (Current Livestock)	$35,000	$5,000				
Decrease in Value per Year	10%	25%				
nd Value (Current Livestock)	$31,500	$3,750	$35,250		Investment	
					Fund	
Cost of New Livestock	$1,500	$3.00	$18,000	<=	$20,000	
End Value (New Livestock)	$1,350	$2.25	$16,200			
Current Livestock	30	2,000				
New Livestock	12	0				
Total Livestock	42	2,000				
	<=	<=				
Barn/House Limits	42	5,000				
Neighboring Farm Work			Neighbor			
	W&S	S&F	Totals			
Wage	$5	$5.50	$10,839			
Hours Worked	782	1,260	2041.8			
Totals	Plantings	Livestock	Neighbor	Total		Available
W&S Hours	97.8	3,120	782	4,000	<=	4,000
S&F Hours	120.4	3,120	1,260	4,500	<=	4,500
Acreage	142	84	0	226	<=	640
Net Income	-$1,840	$44,200	$10,839	$53,199		
End of Year Value		$51,450		$51,450		
Leftover Investment Fund		$2,000		$2,000		
Living Expenses				$40,000		
Total Monetary Worth				$66,649		

Range Name	Cells
AcresPlanted	B7:D7
Available	G39:G41
BarnHouseLimits	B30:C30
CurrentLivestock	B26:C26
HoursWorked	B36:C36
InvestmentFund	F23
MonetaryWorth	E47
NewLivestock	B27:C27
Total	E39:E41
TotalLivestock	B28:C28
Wage	B34:C34

Flood and Early Frost

Plantings				Planting			
	Soybeans	Corn	Wheat	Totals			
W&S Hours Required	1	0.9	0.6	183.3			
S&F Hours Required	1.4	1.2	0.7	219.4			
Net Value	$10	$10	$5	$1,620			
Acres Planted	0	37	250	287			
		>=	>=				
		37	250				
		1	0.05				
		acre/cow	acre/hen				
Livestock			Livestock				
	Cows	Hens	Totals				
Hours Required per Month	10	0.05	620				
Grazing Land Required	2	0	74				
Net Annual Cash Income	$850	$4.25	$52,700				
ning Value (Current Livestock)	$35,000	$5,000					
Decrease in Value per Year	10%	25%					
End Value (Current Livestock)	$31,500	$3,750	$35,250		Investment		
					Fund		
Cost of New Livestock	$1,500	$3.00	$19,500	<=	$20,000		
End Value (New Livestock)	$1,350	$2.25	$16,200				
Current Livestock	30	2,000					
New Livestock	7	3,000					
Total Livestock	37	5,000					
	<=	<=					
Barn/House Limits	42	5,000					
Neighboring Farm Work			Neighbor				
	W&S	S&F	Totals				
Wage	$5	$5.50	$3,567				
Hours Worked	97	561	657.3				
Totals	Plantings	Livestoc	Neighbor	Total		Available	
W&S Hours	183.3	3,720	97	4,000	<=	4,000	
S&F Hours	219.4	3,720	561	4,500	<=	4,500	
Acreage	287	74	0	361	<=	640	
Net Income	$1,620	$52,700	$3,567	$57,887			
End of Year Value		$51,450		$51,450			
Leftover Investment Fund		$500		$500			
Living Expenses				$40,000			
Total Monetary Worth				$69,837			

Range Name	Cells
AcresPlanted	B7:D7
Available	G39:G41
BarnHouseLimits	B30:C30
CurrentLivestock	B26:C26
HoursWorked	B36:C36
InvestmentFund	F23
MonetaryWorth	E47
NewLivestock	B27:C27
Total	E39:E41
TotalLivestock	B28:C28
Wage	B34:C34

f. For the optimal solution obtained under each of the six scenarios (including the good weather scenario considered in parts a-d), calculate what the family's monetary worth would be at the end of the year if each of the other five scenarios occurs instead. In your judgment, which solution provides the best balance between yielding a large monetary worth under good weather conditions and avoiding an overly small monetary worth under adverse weather conditions?

Grandpa has researched what the weather conditions were in past years as far back as weather records have been kept and obtained the data shown on the next page. With these data, the family has decided to use the following approach to making its planting and livestock decisions. Rather than the optimistic approach of assuming that good weather conditions will prevail (as done in parts a-d), the average net value under all weather conditions will be used for each crop (weighting the net values under the various scenarios by the frequencies in the above table).

Scenario	Frequency
Good weather	40%
Drought	20
Flood	10
Early frost	15
Drought and early frost	10
Flood and early frost	5

g. **Modify the linear programming model formulated in part b to fit this new approach.**

Expected (Average) Net Value for Each Crop

Scenario	Frequency	Soybeans	Corn	Wheat
Good weather	40%	70	60	40
Drought	20%	-10	-15	0
Flood	10%	15	20	10
Early frost	15%	50	40	30

Drought and early frost	10%	-15	-20	-10
Flood and early frost	5%	10	10	5

Soybeans:

$=(0.4)(70) + (0.2)(-10) + (0.1)(15) + (0.15)(50) + (0.1)(-15) + (0.05)(10)$
$=28 + (-2) + 1.5 + 7.5 + (-1.5) + 0.5 = 34$

Corn:

$=(0.4)(60) + (0.2)(-15) + (0.1)(20) + (0.15)(40) + (0.1)(-20) + (0.05)(10)$
$=24 + (-3) + 2 + 6 + (-2) + 0.5 = 27.5$

Wheat:

$=(0.4)(40) + (0.2)(0) + (0.1)(10) + (0.15)(30) + (0.1)(-10) + (0.05)(5)$
$= 16 + 0 + 1 + 4.5 + (-1) + 0.25 = 20.75$

New expected values:

Crop	**Updated Expected Net Value**
Soybeans	**$34**
Corn	**$27.50**
Wheat	**$20.75**

New Objective Function:

Maximize $Z = 850C + 4.25H + 34A1 + 27.5A2 + 20.75A3 + (20{,}000 - 1500C - 3H) + 1350C + 2.25H - 40{,}000$

$Z = (850+1350-1500)C + (4.25+2.25-3)H + 34A1 + 27.5A2 + 20.75A3 + 20{,}000 - 40{,}000\ Z$

$Z = 700C + 3.5H + 34A1 + 27.5A2 + 20.75A3 - 20{,}000Z$

Constraints:

Land Constraint: $2C+A1+A2+A3 \leq 640$

Winter/Spring Labor Constraint: $1A1 + 0.9A2 + 0.6A3 + 120C + 0.6H \leq 4000$

Summer/Fall Labor Constraint: $1.4A1 + 1.2A2 + 0.7A3 + 120C + 0.6H \leq 4500$

Feed Constraints:

- Corn acres must cover cows: $A2 \geq C$
- Wheat acres must cover hens: $A3 \geq 0.05H$

Livestock Housing Constraints:

- $C \leq 42$
- $H \leq 5000$

Non-negativity Constraints: $C, H, A1, A2, A3 \geq 0$

h. **Repeat part c for this modified model.**

Microsoft Excel 16.96 Sensitivity Report
Worksheet: [Case Template.xlsx]Case 5-3b
Report Created: 4/26/25 13:05:40

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
B27	New Livestock Cows	12	0	70	1E+30	22.5
C27	New Livestock Hens	0	-0.02	0.35	0.02	1E+30
B7	Acres Planted Soybeans	414	0	21.3	7.5	0.4
C7	Acres Planted Corn	42	0	16.4	4.9	22.5
D7	Acres Planted Wheat	100	0	13.9	0.4	1E+30

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
B28	Total Livestock Cows	42	22.5	42	1.333333333	12
C28	Total Livestock Hens	2000	0	5000	1E+30	3000
C7	Acres Planted Corn	42	-4.9	0	414	42
D23	Cost of New Livestock Totals	$18,000.00	0	20000	1E+30	2000
D7	Acres Planted Wheat	100	-7.4	0	414	100
E39	W&S Hours Total	4000	0	4000	1E+30	0
E40	S&F Hours Total	4500	0	4500	1E+30	0
E41	Acreage Total	640	21.3	640	1E+30	414

The model predicts that the Ploughman family's monetary worth at the end of the coming year will be approximately $80,537. This result reflects an optimized allocation of their farmland, labor, and investment funds, taking into account the historical weather probabilities and the average expected net values for each crop. The plan involves planting 414 acres of soybeans, 42 acres of corn, and 100 acres of wheat, while maintaining a livestock herd of 42 cows and 2,000 hens. The farm also invests in 12 new cows to maximize profitability.

All available land and labor are fully utilized, and the constraints for livestock housing and feed requirements are satisfied. Compared to earlier optimistic scenarios, the expected monetary worth is lower, primarily because the net values of crops have been adjusted downward to reflect the risks of droughts, floods, and early frosts.

i. **Use a shadow price obtained in part h to analyze whether it would be worthwhile for the family to obtain a bank loan with a 10 percent interest rate to purchase more livestock now beyond what can be obtained with the $20,000 from the investment fund.**

The shadow price is zero for the investment so it would not be worthwhile for the family

to obtain a bank loan with 10% interest rate to purchase more livestock beyond the $20,000 from the investment fund, because it would not increase total profit according to the current optimization model.

j. **For each of the three crops, use the sensitivity report obtained in part h to identify how much latitude for error is available in estimating the net value per acre planted for that crop without changing the optimal solution. Which two net values in general terms, an organization faces an uncertain future where any one of a number of scenarios may unfold. Which one will occur depends on conditions that are outside the control of the organization. The organization needs to choose the levels of various activities, but the unit contribution of each activity to the overall measure of performance is greatly affected by which scenario unfolds. Under these circumstances, what is the best mix of activities?**

From the sensitivity report:

CROP	Objective Coefficient	Allowable Increase	Allowable Decrease
Soybeans	21.3	7.5	0.4
Corn	16.4	4.9	22.5
Wheat	13.9	0.4	1E+30

Latitude for Error

- Soybeans: Can increase by 7.5 or decrease by 0.4
- Corn: Can increase by 4.9 or decrease by 22.5
- Wheat: Can increase by 0.4 or decrease by any amount without affecting the optimal solution
- The net value for wheat and soybeans should be estimated more carefully.

Best Mix of Activities: The flood plan offers the best balance between maximizing monetary worth and minimizing risk across unpredictable weather outcomes.Relying only on the good weather plan is risky; exposes the family to significant losses under drought or frost.

k. **Think about specific situations outside of farm management that fit this description. Describe one.**

The best situation outside of the farm management that fits this description is the, and it's faced by different kinds of situations that bring uncertainty to their businesses in fast fashion clothing. Some examples are UNIQLO, ZARA etc. Similar to the farming industry, the fashion industry also needs to address different problems and scenarios like scarcity of materials,seasons and trends.Any kind of business that has same challenges with farm management need to address these various scenarios so that business can cope up and still be profitable.